Sophie Groenwold

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Professional Experience

Oracle, Senior Data Systems Engineer **Sept 2022 - present**

- Develop infrastructure for Oracle Autonomous Database Serverless (ADB-S), focusing on database **high availability, cloning, and cross-region capabilities**
- Design and implement features in Oracle **RDBMS** using **C** and Oracle ADB-S using **PL/SQL**, partnering with engineers across database and broker teams to deliver end-to-end operability
- Diagnose issues in staging, pre-prod and production environments; deliver database patches during on-call rotation
- Integrate **Codex** into development workflows to accelerate codebase exploration, implementation, and debugging

Cross-region Database High Availability

- Developed database-side changes to persist and retrieve pluggable database (PDB) metadata across regions using **cloud-based XML object storage**, supporting high availability during database relocation operations
- Owned cross-region upload infrastructure supporting database availability across **146 production regions**
- Collaborated with broker engineers to define feature requirements and integrated database-side functionality with broker services through cross-tenancy APIs

Staging Tables for Large Data Loads

- Implemented support for staging tables in ADB-S, extending support for optimized large workloads
- Adapted existing database infrastructure to accommodate staging tables' no-logging behavior, including changes to automatic tablespace allocation, truncation, backup and force-logging policies
- Added compatibility checks to existing **failover, switchover, and restore workflows** to ensure these operations continue when a PDB contains staging tables, despite being no-logging

Read-Only Workloads on Standby Databases

- Enabled standby databases to serve **read-only workloads**, beyond the previously supported closed-only state
- Introduced a **connection string modifier parameter** to distinguish read-only standby connections from their primary PDB counterparts, preventing connection ambiguity and **split-brain scenarios**
- Updated standby connection string metadata and implemented parameter state management across database lifecycle transitions to maintain connection consistency

Twitter, *Software Engineer Intern* **June 2021 - Sept 2021**

- Developed a **GraphQL resolver** to expose internal campaign and audit data through structured **JSON** responses
- Contributed backend infrastructure to support the Ads team's transition to an API- and GraphQL-first architecture
- Worked with frontend and backend engineers to define API requirements and coordinate cross-team deliverables

Professional Experience

University of California, Santa Barbara

Bachelor of Science, Major in Computer Science, Minor in History **June 2022**

Publications

Towards Understanding Gender-Seniority Compound Bias in Natural Language Generation.
Proceedings of Language Resources and Evaluation Conference (LREC) 2022. [paper]

Investigating African-American Vernacular English in Transformer-Based Text Generation.
Proceedings of Empirical Methods in Natural Language Processing (EMNLP) 2020. [paper]

Skills

Languages: C, C++, PL/SQL, Python
Technologies: Oracle RDBMS, Distributed Systems, API Design, GraphQL
AI-Assisted Development: ChatGPT, Codex CLI
Tools: Git, Linux, Vim, VS Code, Jupyter Notebook, LaTeX, HTML/CSS
Practices: Agile, Test-Driven Development, Pair Programming